

In situ measurements of wafer bending curvature during growth of group-III-nitride layers on silicon by molecular beam epitaxy

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Abstract

In this work we show how in situ measurements of wafer curvature helps for monitoring the strain relaxation in group-III-nitride layers grown by molecular beam epitaxy on silicon. In particular, we show the efficiency of GaN/AlN interlayers for growing high quality AlGaIn/GaN heterostructures with residual stress that largely compensates the one appearing upon cooling down to room temperature. © 2006 Elsevier B.V. All rights reserved.

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1. Introduction

The heteroepitaxy of group-III-nitrides often leads to problems related to stress due to the lattice mismatch with the substrates and the significant difference in thermal expansion coefficient. The first leads to high densities of threading dislocations and associated residual stress, and the second generates additional stress upon the cooling down to room temperature (RT) after a growth realized between 700 and 1100 °C approximately. In particular, for the growth of Ga(Al)N films on SiC or silicon substrate, the difference in thermal expansion coefficient is responsible for a tensile stress at RT that can generate cracks in the films. Furthermore, the stressed epitaxial layers and the substrate suffer a bending dependent in sign and intensity on the stress. Excessive bending and the presence of cracks hamper the development of devices especially for large substrates where a significant bow can appear. The growth of GaN on silicon is particularly challenging because of the large lattice mismatch (17%) but also the difference in

thermal expansion coefficients between GaN and Si that is estimated $\alpha_{\text{GaN}} - \alpha_{\text{Si}(111)} = 2.9 \times 10^{-6} \text{ K}^{-1}$; such a thermal mismatch is responsible for a tensile strain of 0.23% at RT if the growth of a totally relaxed GaN film is achieved at 800 °C. As a consequence, without stress engineering schemes, no thick, high-quality GaN layer can be obtained without cracks on Si substrates. The use of AlN interlayers or AlGaIn stress mitigating layers allows the growth of thick high quality GaN films on silicon [1–4]. Indeed, the 2.6% lattice mismatch between GaN and AlN allows GaN to be grown compressively strained and even if a major part of this strain is relaxed, to keep a residual stress capable of compensating the thermal mismatch stress upon cooling down. Nevertheless, the efficiency of such a scheme is very dependent on growth conditions and preceding layer quality, and in situ monitoring of stress is a very useful tool for optimizing such layers. In particular, as shown for the first time in this work in a molecular beam epitaxy (MBE) reactor, the measurement of the curvature of silicon wafer bending during growth of AlN and GaN layers provides very useful information on strain relaxation for the design of structures based on thick GaN buffer layers such as high electron mobility transistors (HEMTs).

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2. Experimental procedure

In the present work, AlGaIn/GaN heterostructures have been grown in a reactive MBE system using ammonia (Riber Compact 21) [5]. We grew heterostructures on 50 mm highly resistive ($\rho > 10\,000\,\Omega\text{ cm}$) Si(111) substrates after thermally removing the silicon oxide at 950°C . A custom-made LAYTEC epicurve curvature measurement system based on two parallel laser beams ($\lambda \sim 635\text{ nm}$) was used, in combination with an EpiTT reflectivity and true wafer temperature measurement system. All measurements were performed with light beams nearly perpendicular to the surface in the center region of the wafer, and thanks to a weak wobble, they could be done while rotating the wafer (8–10 rpm).

As usual, the surface temperature of the wafer was additionally measured using an Ircon infrared pyrometer ($\lambda = 0.91\text{--}0.97\,\mu\text{m}$) calibrated with the $(1 \times 1)\text{--}(7 \times 7)$ surface reconstruction change at 830°C . It is worth noticing that the TT true wafer temperature in addition to some 20°C absolute difference also does not suffer the $\pm 20^\circ\text{C}$ oscillations measured with the Ircon infrared pyrometer due to the implemented real-time emissivity correction. In the TT wafer temperature measurement parasitic offset contributions due to IR radiation out of the hot sources has been reduced to only $\sim 2\text{ K}$ by a specially designed optical head.

In the in situ curvature measurement some data points could not be represented or have been set constant due to zero reflectance of the laser beams for certain GaN thicknesses. This was necessary because growing GaN periodically acts as a perfect anti-reflecting layer on silicon. The bow of as grown samples was measured also ex situ at RT for reference using a surface profiler along the diameter of each wafer.

3. Results and discussion

The growth can be summarized as follows but more details about growth by ammonia MBE of GaN on Si(111) can be found elsewhere [2]. A 40 nm thick AlN layer is first nucleated while the substrate temperature is increased from 600 to 920°C , followed by a $0.25\,\mu\text{m}$ GaN and $0.25\,\mu\text{m}$ AlN stack grown at 800 and 920°C , respectively. This patented layer structure allows the subsequent growth of crack free $2\text{--}3\,\mu\text{m}$ thick GaN. A low growth rate ($0.1\,\mu\text{m/h}$) is used to maintain smooth AlN surfaces, whereas GaN is grown at about $0.6\,\mu\text{m/h}$.

Fig. 1 shows the wafer temperature and the curvature measured during the growth and upon cooling down to RT; this structure contains a $1.7\,\mu\text{m}$ GaN buffer layer capped with a 1 nm AlN/21 nm AlGaIn barrier/5 nm GaN as HEMT active layers. As for many MBE systems the substrate is not in contact with the heater and is heated by radiations; contrary to what happens during MOCVD growth [6], the bending of the silicon substrate does not change the temperature, even for values close to 0.3 m^{-1} .

This is a significant MBE advantage for the growth of temperature sensitive indium containing structures. Once the growth of the first AlN layer is initiated ($t = 3300\text{ s}$, $T = 940^\circ\text{C}$), one can notice the positive curvature that saturates to 0.02 m^{-1} , before the growth of the $0.25\,\mu\text{m}$ GaN layer ($t \sim 5500\text{ s}$, $T = 790^\circ\text{C}$) quickly annihilates the curvature to almost 0 m^{-1} . The $0.25\,\mu\text{m}$ AlN layer ($t \sim 7000\text{ s}$, $T = 900^\circ\text{C}$) reestablishes a curvature that rises from 0.02 to 0.04 m^{-1} .

Despite growth conditions identical to those of the first $0.25\,\mu\text{m}$ GaN interlayer, the $1.7\,\mu\text{m}$ thick GaN buffer rapidly develops a negative curvature that slowly saturates to -0.24 m^{-1} after the deposition of about $1.6\,\mu\text{m}$ of GaN. The growth of the tensile strained AlGaIn barrier layer also contributes to this saturation, before the cooling down to RT reduces the curvature to -0.04 m^{-1} .

The resulting bow b of the 50 mm wafer at RT can be simply calculated using Eq. (1):

$$b = 1/\kappa \cdot (1 - \cos(W \cdot \kappa / 2)). \quad (1)$$

W is the wafer diameter, and κ the curvature. The strain ε can be evaluated in first approximation by the Stoney formula [7]:

$$\kappa = 6M_f \cdot \varepsilon \cdot h_f / M_s \cdot h_s^2. \quad (2)$$

M and h are the biaxial module and thickness, subscripts f and s referring to the grown film and to the substrate, respectively. However, this linear approximation is only valid for $h_f/h_s \ll 1$ and for small values of stress; more accurate calculations have been developed in a non-linear theory for large bending [8].

Table 1 summarizes the curvatures measured at high temperature ($\sim 800^\circ\text{C}$) at the end of growth and at RT for a set of sample we have grown with the same layer structure but with varying growth conditions, namely the ammonia flow rate and the temperature for the growth of the thick GaN buffer layer. The RT bow calculated from the in situ data with Eq. (1) is reported as well as the bow measured ex situ with the surface profiler.

First, the consistency of RT measured in situ bows with ex situ measured bows is obvious and attests to the accuracy of the measurements. The mean error for bow measurement is less than $3\,\mu\text{m}$ and corresponds to a curvature of 0.01 m^{-1} which is within the precision of ex situ profiler curvature measurements as evaluated on bare silicon substrates. Second, for most of the samples, a similar behavior is observed for the development of the curvature during the growth of the thick GaN layer, and the difference of curvature between growth temperature and RT is almost constant (0.2 m^{-1}). This value is 20% lower than the one calculated using the Stoney formula (Eq. (2)) with parameters for GaN and Si substrate ($M_f/M_s = 1.7$, $h_f = 1.7\,\mu\text{m}$, $h_s = 380\,\mu\text{m}$, $\varepsilon = 0.0023$), but it reasonably fits using a 0.7 correction coefficient calculated from the non-linear theory of curvature [8].

Except for two samples, all wafers present a negative (convex) curvature, attesting for a residual compressive

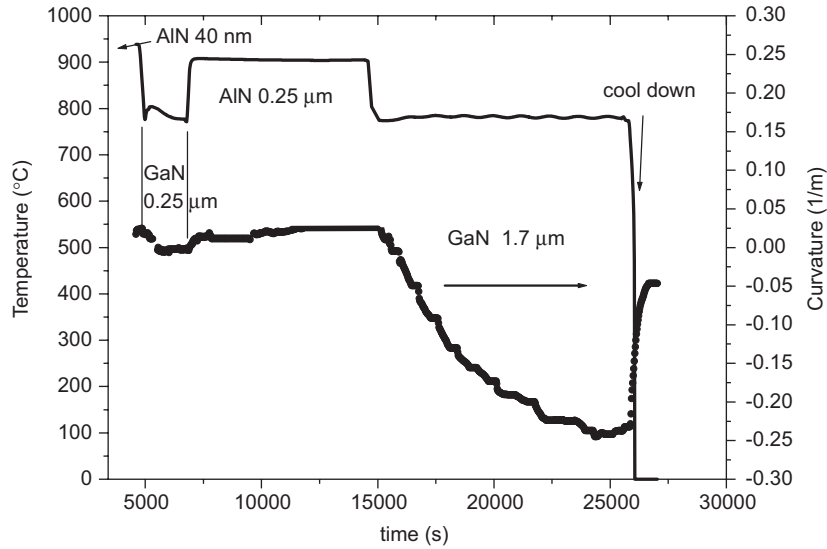


Fig. 1. Curvature and true wafer temperature measured during the growth of the reference sample PTC 240.

Table 1

High temperature (HT) and room temperature (RT) curvatures and bow measured on the samples

Sample	HT curvature (m^{-1})	RT curvature (m^{-1})	HT–RT curvature (m^{-1})	HT calculated bow (μm)	RT calculated bow (μm)	RT bow (μm)
PTC239	−0.27	−0.065	−0.205	−84	−20	−29
PTC240	−0.24	−0.04	−0.2	−75	−12	−17
PTC243	−0.19	0.01	−0.2	−59	3	2
PTC245	−0.27	−0.035	−0.235	−84	−11	−14
PTC246	−0.08	0.1	−0.18	−25	31	34
PTC247	−0.29	−0.09	−0.2	−91	−28	−31
PTC248	−0.26	−0.06	−0.2	−81	−19	−18
PTC249	−0.31	−0.058	−0.252	−97	−18	−25
PTC251	−0.254	−0.047	−0.207	−79	−15	−17

strain. The present results show the high efficiency of this specific AlN–GaN growth procedure to generate a total stress that compensates the one appearing upon cooling down. However, this efficiency is clearly dependent on the quality of each buried layer. From Eq. (2), any increment of material deposited with a fixed strain state produces a linear change of curvature. On the contrary, any non-linear variation indicates a change of strain in the grown material. As shown in Fig. 1, the first GaN interlayer grown with a very high density of dislocations (several 10^{10} cm^{-2}) only compensates the effects of the AlN nucleation layer, whereas for the same deposited thickness ($0.25 \mu\text{m}$) the second GaN layer clearly develops a noticeable negative curvature while the dislocation filtering by interactions followed by bending results in densities in the low 10^{10} cm^{-2} and falls down to $5 \times 10^9 \text{ cm}^{-2}$ as has been checked by plan-view transmission electron microscopy. The same conclusion can be drawn for AlN layers that are grown with a high defect density so that they quickly relax

the mismatch strain (after growth of $\sim 100 \text{ nm}$) and produce in total only a limited positive curvature.

Furthermore, the development of the curvature for the thick GaN layer is observed to depend on the growth conditions. Fig. 2 shows the curvature of the sample PTC246 grown using the same conditions as PTC240 for AlN and GaN buried layers, but with a 20°C higher substrate temperature for the thick GaN buffer. Despite similar RHEED patterns indicating a two-dimensional growth mode and curvature quickly developing towards negative values, the saturation of curvature at a low value -0.08 m^{-1} can be observed after only half of the GaN growth. This type of structure cannot compensate the thermal mismatch and results in a $+0.1 \text{ m}^{-1}$ curvature (concave) wafer with cracks. A higher threading dislocation density (about twice) on its surface confirms the primary role of GaN-on-AlN nucleation conditions on the defect density and relaxation rate. Correlation of growth stress and dislocation behavior has been recently reported for the growth by MOCVD of GaN on thick AlGaN on Si(111) [9]. In a similar way, cross-sectional transmission electron microscopy imaging is in progress to identify the mechanism involved for strain relaxation at a faster rate, i.e. nucleation, interactions, or bending of dislocations with Burgers vector having an edge component parallel to the interfaces.

4. Conclusion

The growth of high quality group-III-nitride layers on silicon necessitates understanding of the development of stress towards strain relaxation in multilayered structures and management of thermal mismatch stress upon cooling down. In situ measurements of the substrate curvature help to understand and to monitor the stress in order to achieve

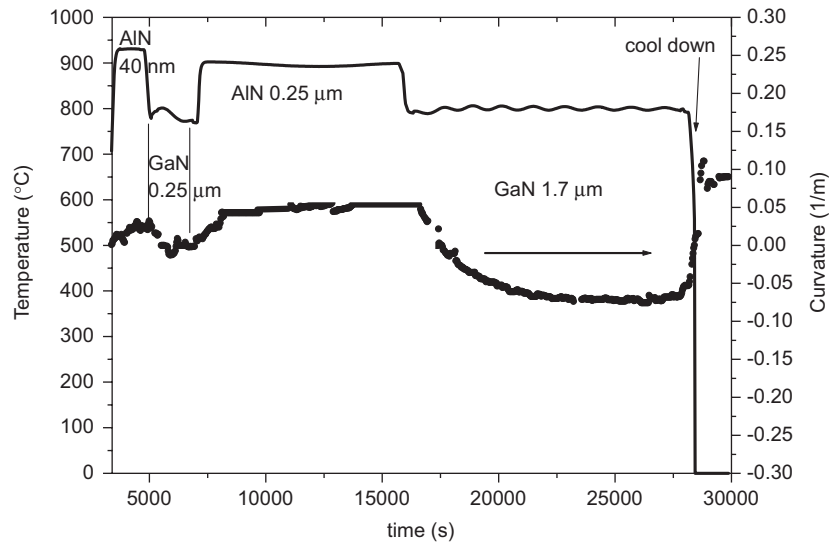


Fig. 2. Curvature and true temperature measured during the growth of the sample PTC 246.

high quality–low curvature heterostructures as shown here with AlGaIn/GaN HEMTs grown by MBE.

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